Unit Vectors

SYNOPSIS

You have already written one. In Note 3 the launched ball's velocity came out as $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$, and those two hatted symbols did real work in an equation that carried the whole argument — without ever being defined. This note names them. A unit vector is **pure direction**: magnitude exactly 1, and no units at all. Everything with a size and a unit sits in front of it. Once you have $\hat{\imath}$ and $\hat{\jmath}$, every vector in the plane is two ordinary numbers, and adding vectors becomes collecting like terms.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- **PE1** **State what makes a vector a unit vector** — magnitude exactly 1, no units — and explain why the hat contributes nothing to the units of the quantity it describes.
- **PE2** **Write a vector in unit-vector form** from its components, and read a unit-vector expression back as a magnitude and a direction.
- **PE3** **Multiply a vector by a scalar** and say what changes, what cannot change, and why a negative scalar is a statement about direction rather than a negative magnitude.
- **PE4** **Add two vectors given in unit-vector form** by collecting the $\hat{\imath}$ and $\hat{\jmath}$ terms separately, then recover the magnitude and direction of the result.

Predict first. In the velocity you wrote in Note 3, $\vec{v}(t) = v_{0x}\,\hat{\imath} + (-9.8t)\,\hat{\jmath}$, what does $\hat{\jmath}$ contribute to the *units* of $\vec{v}$?

- Metres per second.
- Metres.
- Seconds.
- Nothing at all — $\hat{\jmath}$ has no units.

You already used this

Look again at the equation from Note 3 and read it in pieces rather than as one lump:

$$\vec{V}(t) = v_{0x}\,\hat{i} + (-9.8\,t)\,\hat{j}$$

What each piece is doing: v_{0x} is a number with units (m/s) and it is the whole *size* of the horizontal part. $\hat{i}$ says *which way* that part points, and contributes no size and no units. Likewise $-9.8t$ is a number with units (m/s), and $\hat{j}$ says it points along the vertical axis — with the minus sign meaning downward, because up was chosen as positive.

So the equation is not one complicated object. It is **two ordinary numbers**, each with a label saying which axis it belongs to. That is the whole idea, and the rest of this note just makes it official.

What makes a vector a *unit* vector

📐 Three things, and that is the whole definition

1️⃣
Magnitude exactly 1
Not approximately, not "about one unit long." Exactly 1, always, in every unit system.

🧭
Along a coordinate axis
$\hat{i}$ points along +x, $\hat{j}$ along +y. They are locked to your axes and do not rotate with the object.

🚫
No units of its own
It is dimensionless. All the physics, and every unit, lives in the number in front.

$$|\hat{i}| = |\hat{j}| = 1$$

Where this holds: always — it is the definition, not a result, and it does not depend on the units of the grid you drew. A consequence worth keeping: a **magnitude is never negative**. When you write $-4\,\hat{i}$, the minus sign is a statement about *direction* — magnitude 4, pointing along −x. "The magnitude is −4" is always wrong.

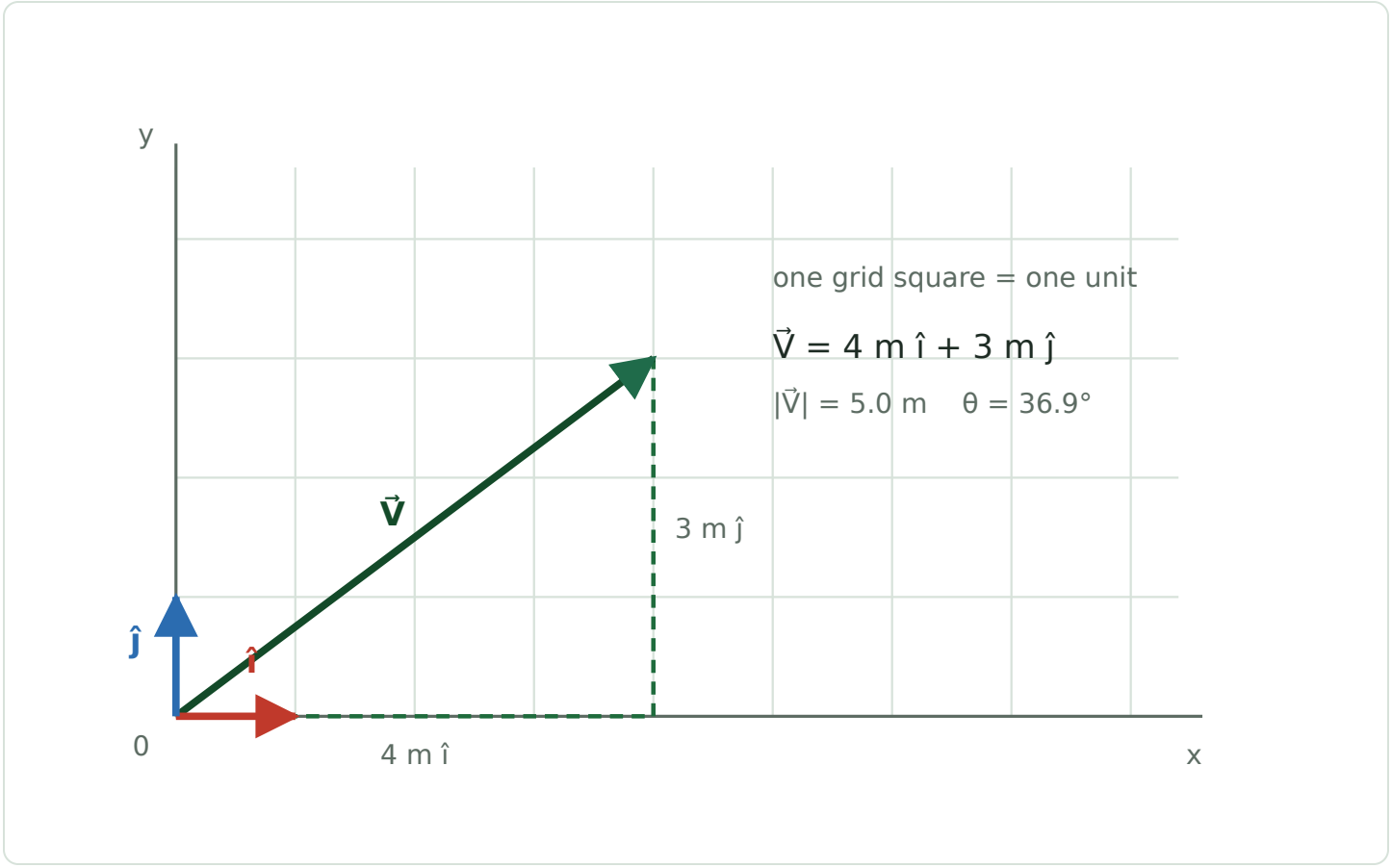


Figure 1. The red and blue arrows are each exactly one square long and locked to an axis — those are î and ĵ. The green arrow is free to be any length in any direction, and it is built out of the other two: four metres' worth of î, then three metres' worth of ĵ. The 3–4–5 triangle you already know does the rest.

What a number in front can — and cannot — do

Multiplying a vector by a scalar does exactly two things: it changes the **length**, and if the scalar is negative it **reverses** the direction end for end. That is the complete list. **No scalar can rotate a vector** to a new angle. If you need a direction that is not along an axis, you build it by adding î and ĵ pieces — which is the next section, and is exactly what Note 2 had you doing with a ruler.

Adding vectors becomes collecting like terms

This is the payoff, and it is the reason the notation exists. In Note 2 you discovered that components add independently: $C_x = A_x + B_x$ and $C_y = A_y + B_y$. Written with hats, that discovery becomes ordinary algebra — gather the î terms, gather the ĵ terms, and do not let them mix:

$$\vec{A} = (3.0\ \text{m})\,\hat{\imath} + (5.0\ \text{m})\,\hat{\jmath} \qquad \vec{B} = (-1.0\ \text{m})\,\hat{\imath} + (2.0\ \text{m})\,\hat{\jmath}$$
$$\vec{A} + \vec{B} = (3.0 - 1.0)\ \text{m}\,\hat{\imath} + (5.0 + 2.0)\ \text{m}\,\hat{\jmath} = (2.0\ \text{m})\,\hat{\imath} + (7.0\ \text{m})\,\hat{\jmath}$$

Where this holds: both vectors must be the same kind of quantity, in the same units, on the same set of axes. You may add a displacement to a displacement; you may not add a displacement to a velocity, however tidy the algebra looks. **The $\hat{\imath}$ and $\hat{\jmath}$ terms never mix** — that is the independence of the axes, the same fact Note 3 used to argue that horizontal motion cannot change the fall time.

And the magnitude and direction come back the way they always have, from Note 2a:

$$|\vec{A} + \vec{B}| = \sqrt{(2.0)^2 + (7.0)^2}\ \text{m} = 7.3\ \text{m} \qquad \theta = \tan^{-1}(7.0 / 2.0) = 74.1^\circ$$

Watch the quadrant. The inverse tangent alone cannot tell +x, +y from −x, −y — it returns the same angle for both. Sketch the vector first and read the quadrant off the *signs* of the two components, exactly as you did in Note 2a.

Two notations, same three arrows

YOU WILL SEE	READ AS	WHERE
$\hat{\imath}, \hat{\jmath}, \hat{k}$	"i hat, j hat, k hat"	**This course.** It is what your textbook and WebAssign use, and it is what is written on every note from here on.
$\hat{x}, \hat{y}, \hat{z}$	"x hat, y hat, z hat"	The same three vectors, named after the axes instead. You will meet it in the Chapter 3 reference pages and in plenty of other books. $\hat{\imath} = \hat{x}$, $\hat{\jmath} = \hat{y}$, $\hat{k} = \hat{z}$.

Neither is more correct. Pick one and stay with it inside a single problem — mixing them halfway through a solution is how sign errors get in.

> **Looking ahead —** in **Note 4** the launcher tilts, and the velocity picks up a vertical piece at the start: $\vec{v}(t) = v_0\cos\theta\,\hat{\imath} + (v_0\sin\theta - 9.8t)\,\hat{\jmath}$. Every term in it is something you can now read. The notation is not decoration — it is what keeps the two motions visibly separate while they share one clock.

🎟 Exit ticket

Four quick questions — check yourself before you close this note.

Q1. **The magnitude of $\hat{\imath}$ is…**

- ○ 1 metre.
- ○ Exactly 1, with no units.
- ○ Whatever one grid square represents.
- ○ Undefined until you choose axes.

Q2. **A student writes "the magnitude of $-4\,\hat{\imath}$ is -4." What is wrong?**

- ○ Nothing — that is correct.
- ○ The magnitude is 4; the minus sign gives the direction, along $-x$.
- ○ The magnitude is 4 and the vector points along $+x$.
- ○ You cannot multiply a unit vector by a negative number.

Q3. **$\vec{A} = (6.0\text{ m})\,\hat{\imath} + (2.0\text{ m})\,\hat{\jmath}$ and $\vec{B} = (-2.0\text{ m})\,\hat{\imath} + (5.0\text{ m})\,\hat{\jmath}$. What is the $\hat{\jmath}$ component of $\vec{A} + \vec{B}$, in metres?**

_____ m

Q4. **Multiplying a vector by the scalar 3 will…**

- ○ Rotate it by 3 degrees.
- ○ Make it three times as long, same direction.
- ○ Make it three times as long and reverse it.
- ○ Change its units.

Check my answers | Clear

Your notes

BEFORE CLASS

Guided notes — blank (PDF)

Print it and build the vectors on the grid as you follow along.

GO DEEPER

Chapter 3 reference — Unit Vectors

The full slide treatment, with more worked examples and the three-dimensional case.

Copyright © 2026 by Dr. Byungmoon Cho · FLCC Physics · PHY 151